2024

1- $y'' + 2y' + y = \cos$

$y(0) = 0$

$y'(0) = 1$

$y'' + 2y' + y = 0$

$x^2 + 2x + 1 = 0$

$= (x - (-1))^2 \cdot \quad y_h(x) = ae^{-x} + bxe^{-x}$

$A\cos x + B\operatorname{sen} x$

$(A\cos x + B\operatorname{sen} x)'' + 2(A\cos x + B\operatorname{sen} x) + Ax + Bx$

$= -A\cos x - B\operatorname{sen} x - 2A\operatorname{sen} x + 2B\cos x + Ax + Bx$

$= \cos x\,(-A + 2B + A) + \operatorname{sen} x\,(-B - 2A + B) = x$

$= 1 \iff B = \frac{1}{2} \qquad \iff A = 0$

$\to \frac{1}{2}\operatorname{sen} x = y_p(x)$

$\left\{ \lambda e^{-x} + \mu x e^{-x} + \frac{1}{2}\operatorname{sen} x \right\}_{\lambda,\mu \in \mathbb{R}}$

$\begin{cases} y(0) = 0 \\ y'(0) = 1 \end{cases} \iff \begin{cases} 0 = \lambda + 0 + 0 \\ 1 = \mu e^{-0} + \mu 0 e^{-0} + \frac{1}{2}\cos(0) \end{cases}$

$1 - \frac{1}{2} = \mu$

$y(x) = \frac{1}{2} x e^{-x} + \frac{1}{2}\operatorname{sen} x$

$y(\pi) = \frac{1}{2}\pi e^{-\pi} + 0$

$\cdot \pi$

c

3-